Week 6 Notes & Quiz

Notes for all five Week 6 lectures, made only from the 150 slides in *WEEK 6.pdf*. Each yellow tag such as S29 is the slide (PDF page) the point comes from. The 50-question practice questions and answer key are at the end.

STOP	LECTURE	SLIDES
L26	**Supply Chain Uncertainty and Processes**	S1–S25
L27	**Supply Chain Operational Views**	S26–S53
L28	**Supply Chain Models and Modelling Systems**	S54–S88
L29	**Supply Chain Drivers: facilities, inventory, transportation**	S89–S120
L30	**Supply Chain Drivers: information, sourcing, pricing**	S121–S150
+	**Cheat sheet · Slide quirks · Quiz (50 Qs)**	—

LECTURE 26 S1–S25

Supply Chain Uncertainty and Processes

- Hook: are you 100% sure the 1,500-unit truck arrives this weekend? "Too much inventory kicks off finance; not enough inventory takes off your customers." S3
- **Supply chain uncertainty**: unpredictability and volatility that disrupt the flow of goods, from natural disasters, geopolitics, labour shortages, logistics disruptions, demand swings. S4

Demand vs implied demand uncertainty

- **Demand uncertainty**: overall uncertainty of customer demand (trends, behaviour, economy). S5
- **Implied demand uncertainty**: uncertainty for the portion of demand the supply chain **plans to satisfy**, given order frequency, lead times and service levels. S5
- A firm serving **emergency orders** faces higher implied uncertainty than one supplying the same product with long lead time. **Raising the service level raises implied demand uncertainty**, even if demand uncertainty is unchanged. S6

Supply uncertainty

- Driven by **supply chain capability** (e.g. low yields with new PC components) and the **life-cycle position** of the product: new phones have high supply uncertainty, salt in a supermarket low. S7–S8

Countering uncertainty

Diversify suppliers · enhance visibility · build flexibility (buffer stock, agile manufacturing) · strengthen relationships · risk management practices · invest in resilience · monitor and adapt; plus automation, AI and analytics. S9–S11

> UNCERTAINTY VS RISK S13
>
> **Risk** is a function of outcome and probability, so it **can be estimated**. **Uncertainty** occurs when decision makers **cannot estimate** the outcome of an event or its probability.

TYPE (MICHEL WANG, AUSTRALIAN COURIER INDUSTRY)	SOURCES S14–S16
Logistics uncertainty and risk	Delivery delays, transport network, storage, carrier strength, machine failures
Information uncertainty and risk	Unavailable or delayed information, infrastructure breakdown; accuracy, availability, timeliness
Customer-related	Consignors and consignees: volatile behaviour, reputation, receivables, forecast errors
Environment	PESTEL-type factors: labour, congestion, disasters, fuel prices, regulation

Mitigating uncertainty through automated planning 5 items

- **Regular plan generation**: moving plan reviews from yearly to weekly can cut inventory by 37%. S17
- Consistency in planning; integration of more data sources; visibility across the chain (reduces the **bullwhip effect**); holistic KPI tracking. S18–S19

Supply chain macro processes

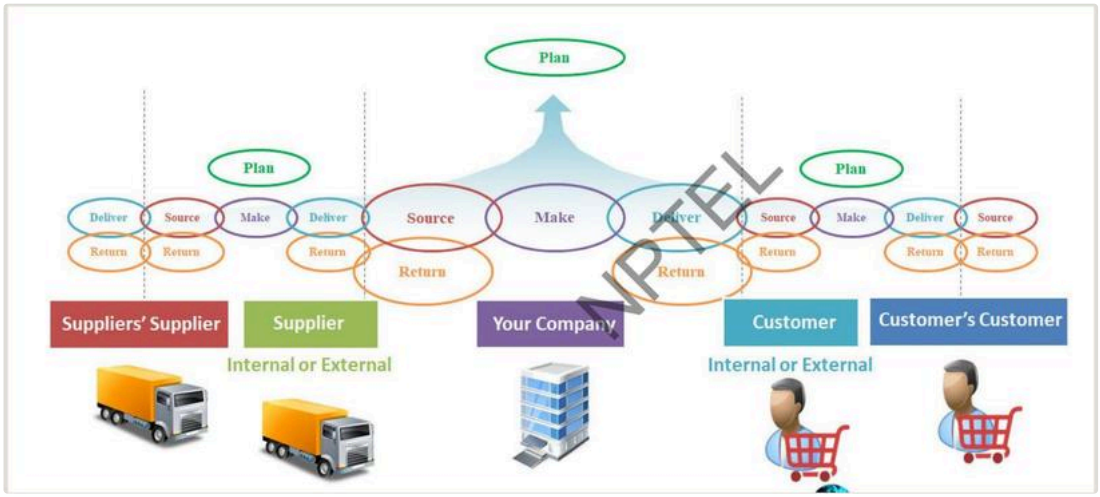
MACRO PROCESS	INTERFACE · AIM · PROCESSES S20–S21
CRM	Firm ↔ customers. Generate demand, facilitate ordering and tracking. Marketing, pricing, sales, order management, call centre.
ISCM	Internal to the firm. Fulfil demand efficiently. Capacity planning, demand and supply planning, order fulfilment.
SRM	Firm ↔ suppliers. Arrange supply sources. Supplier evaluation and selection, term negotiation, communication.

LECTURE 27 S26–S53

Supply Chain Operational Views

SCOR model

- **Supply Chain Operations Reference (SCOR)** model: the planning and diagnostic system of SCM, from the **Supply-Chain Council** (a global non-profit consortium). S29
- Five processes: **Plan, Source, Make, Deliver, Return**. Five performance dimensions: **reliability, responsiveness, agility, costs, asset management efficiency**. S29–S30



Original slide S31. Plan, source, make, deliver and return link suppliers' supplier → supplier → your company → customer → customer's customer.

- Elements, the **4Ps**: **Performance** (KPIs), **Process** (plan, source, make, deliver, return, enable), **Practice** (methods and tools), **People** (skills, training). S32–S33
- More than **250 metrics**, summarized as reliability (right quantity and condition), responsiveness (speed), costs, asset management (working capital, fixed assets). S34
- Advantages: whole-chain scope, benchmarking, metric alignment across boundaries, common vocabulary, standardized improvement. S35
- Disadvantages: complexity, one-size-fits-all, lack of flexibility (linear), limited scope (no social responsibility or sustainability), dependence on data. S36
- Example: e-commerce, where digital processes are well documented and easy to analyse with SCOR. S37–S38

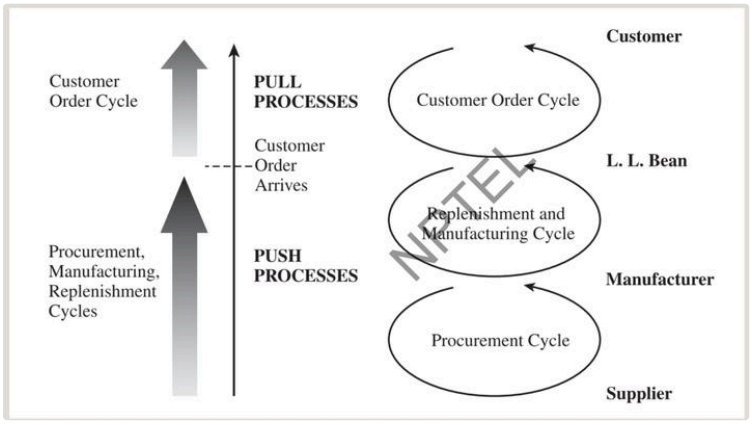
Cycle view

- Processes split into **cycles**, each at the interface between two successive stages; defines roles and owners. Useful for **operational** decisions. S39 S41

CYCLE	INTERFACE	STEPS S44–S47
Customer order	Customer – retailer	Customer arrival → order entry → order fulfilment → order receiving
Replenishment	Retailer – distributor	Retail order trigger → entry → fulfilment → receiving
Manufacturing	Distributor – manufacturer	Order arrival → production scheduling → manufacturing and shipping → receiving
Procurement	Manufacturer – supplier	Order per production schedule → supplier scheduling → component manufacturing and shipping → receiving

Push/pull view

- **Pull**: executed in response to a customer order (**reactive**). **Push**: executed in anticipation of orders (**speculative**). S48



Original slide S49, L.L. Bean. The customer order cycle is **pull**; replenishment, manufacturing and procurement cycles are **push**. The push/pull boundary is where the customer order arrives.

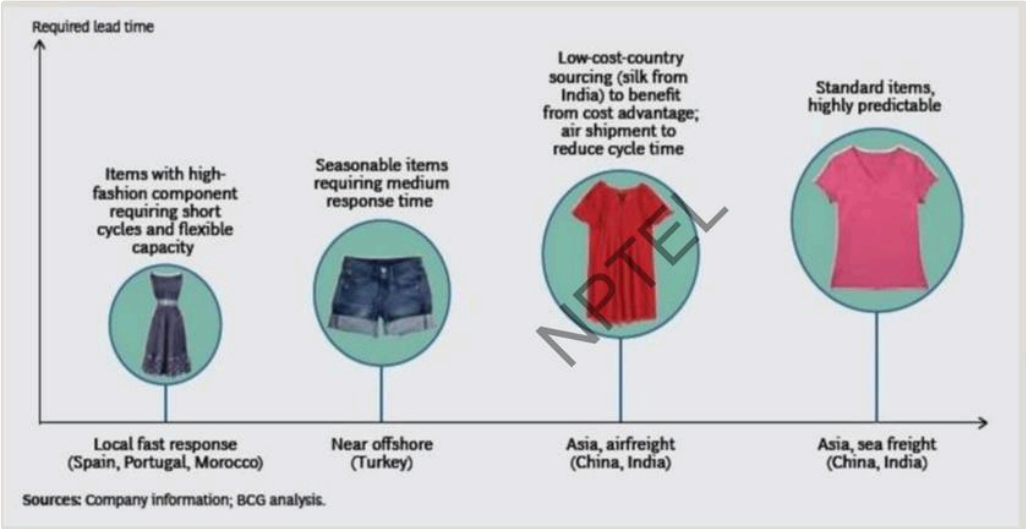
LECTURE 28 S54–S88

Supply Chain Models and Modelling Systems

- **Supply chain modelling**: designing and analysing a supply chain to find improvements and simulate scenarios. Models include **static, dynamic and Monte Carlo** simulation. Benefits: lower costs, efficiency, customer satisfaction. S56–S57
- 8 major components: **inventory** (the most critical, hits the bottom line), transportation, warehousing, manufacturing, procurement, distribution, customer service, IT. S58–S59
- Model types for decisions: **linear programming** (optimize resource distribution), simulation, queuing, statistical. Choose by the optimization goal. S60

Types of supply chain models

MODEL	KEY IDEA · EXAMPLE
Continuous	Steady flow, JIT, stable demand; less flexible. **Amazon**. S62–S63
Fast	Quick delivery, short life cycles. **Zara** (new trends within weeks vs six months). S65
Efficient chain	Reduce waste in highly competitive markets; relies on accurate production predictions; vulnerable to labour or material shortages. S66–S67
Agile	Responsive to change; specialist goods; profitable only up to a volume threshold. Four traits: **process alignment, virtual integration, network base, market sensitivity**. **Nike**. S69–S70
Custom-configured	Customized to customer demand; mixes agile and continuous flow. **Dell**. S71
Flexible	Handles peaks and long low periods; needs part segmentation, precise stocking algorithms, flexible planning. **Hewlett Packard**. S72
Virtual (future)	Cloud and analytics. **Procter & Gamble**. S73



Original slide S68 (BCG). Zara by segment: high-fashion items → local fast response (Spain, Portugal, Morocco); seasonal items → near offshore (Turkey); low-cost sourcing → Asia airfreight (China, India); standard predictable items → Asia sea freight.

- Best practices: understand needs, choose the approach, validate with accurate data, test "what if" scenarios, consider several models, involve stakeholders, update, document assumptions, train, monitor. S74–S75
- Challenges: complex relationships, imperfect data, multiple objectives, dynamic environment, integration, resource constraints, scenario complexity, risk, change management. S76–S77
- **Supply chain control tower**: a cloud-based solution using ML, AI and IoT for end-to-end, real-time visibility; the future of SC modelling. S78

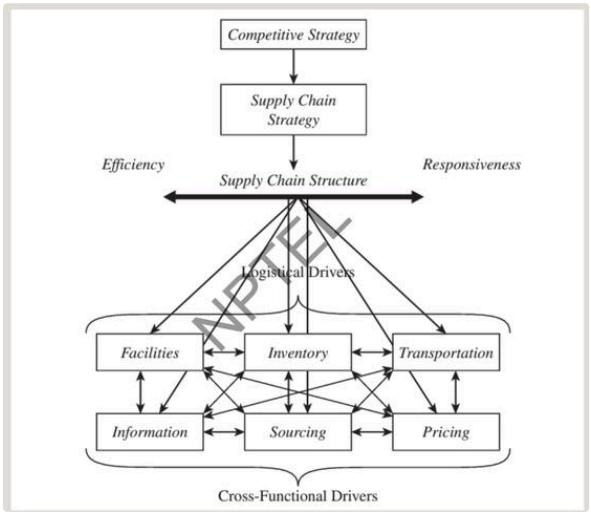
Case studies

- **Intel Atom**: supply chain cost about **$5.50 per chip** was fine for $100 chips but not for a **$20** chip. Only lever: **inventory**. Switched to **make-to-order** (pilot in Malaysia), cut the test window from 5 days to a biweekly 2-day process, added S&OP and VMI. Order cycle fell from **nine weeks to two**; cost cut by more than **$4 per unit**. S80–S82
- **Chemical company**: fragmented network after acquisitions; a large **MILP** (mixed-integer linear programming) model → **10% lower costs** and **next-day delivery for 95%** of products. Hurdles: data across ERP systems, scepticism. S83–S85

LECTURE 29 S89–S120

Supply Chain Drivers: Facilities, Inventory, Transportation

- Supply chain drivers shape **responsiveness and efficiency**: **facilities, inventory, transportation, information, sourcing, pricing**. S91–S93



Original slide S95 (Chopra & Meindl). Competitive strategy → supply chain strategy → structure (efficiency vs responsiveness). **Logistical drivers**: facilities, inventory, transportation. **Cross-functional drivers**: information, sourcing, pricing.

- Cross-functional drivers have broader scope, rely on internal collaboration, and impact **responsiveness and agility** more. Logistics drivers focus on physical movement, external partners, **efficiency and cost**. S96–S97

Facilities

- Physical places where inventory is stored or transformed. **Centralizing** raises efficiency but cuts responsiveness; decentralizing does the reverse. S98
- Toyota and Honda open plants in every major market for responsiveness; Honda plants could build SUVs and cars, staying busy in the 2008 downturn. S99
- Decisions: role (flexible vs dedicated capacity; product vs functional focus; cross-docking vs storage), location, capacity (excess capacity = flexible but costly). S100
- Metrics: capacity, utilization, processing/setup/down/idle time, cost per unit, quality losses, theoretical and actual flow time, **flow time efficiency** (theoretical ÷ actual), product variety, volume share of top 20% SKUs, batch size, production service level. S101–S102

Inventory

- Role: matches supply and demand, exploits **economies of scale**, increases responsiveness, but high levels raise costs. S103–S104
- **Little's law** links inventory (I), flow time (T) and throughput (D): $I = D \times T$. Less inventory at the same throughput means shorter flow time. S104
- Amazon: best-sellers in many regional warehouses; slow movers in few; slowest printed on demand or ordered from the publisher. S106
- Components: level of product availability, **cycle inventory** (usual demand), **safety inventory** (buffer), **seasonal inventory** (peaks). S107
- Metrics: cash conversion cycle, average inventory, seasonal inventory, inventory turnover (COGS or sales ÷ average inventory), **DIO** (average inventory ÷ COGS per day), batch size, average safety inventory, fill rate, fraction of time out of stock, obsolete inventory. S108–S110

Transportation

- Faster modes improve responsiveness but cost more. **Dell** flies components from Asia to keep inventory low. S112–S113
- **Blue Nile**: FedEx overnight delivery plus centralized diamond inventory, no physical stores. S114
- Decisions: network design (direct or via consolidation points), mode (air, truck, rail, sea, pipeline, internet). Inbound transport hits COGS; outbound hits SG&A. S115
- Metrics: average inbound/outbound transport cost, shipment size, cost per shipment, fraction transported by mode. S116–S117
- Trade-off: air/express = responsive; sea/rail = efficient. S118

LECTURE 30 S121–S150

Supply Chain Drivers: Information, Sourcing, Pricing

Information

- Role: asset utilization, coordination of flows, responsiveness, lower costs. Examples: **Seven-Eleven Japan**, **Wal-Mart**. S123–S125
- Share the **minimum information necessary**; beyond a point, marginal cost exceeds marginal benefit. S127
- **Sunsweet Growers** (dried fruit, California) replaced Excel with an S&OP suite: production overruns fell from **30% to under 15%**, forecast accuracy up **15–20%**, warnings **2–3 weeks** earlier. S128
- Decisions: push vs pull, coordination and sharing, **S&OP** (align sales forecasts with production and inventory), enabling technologies (ERP, analytics, cloud). S129
- Metrics: forecast horizon (≥ decision lead time), update frequency, forecast error, seasonal factors, variance from plan, **ratio of demand variability to order variability** (below 1 suggests a **bullwhip effect**). S130–S131
- Trade-off: **complexity vs value**; sharing aggregate sales may beat detailed POS data. S132

Sourcing

- Buying goods and services; deciding responsive vs efficient sources, in-house vs third party. IKEA sources low-cost modules; Dell sources some PCs from China. S133
- Outsource next-day delivery to package carriers; Zara keeps quick response in-house. S134
- **Cisco**: outsources almost all manufacturing. Low-end home routers made in **China** for efficiency; high-end products from US contract manufacturers for responsiveness. S135
- Decisions: in-house or outsource, supplier selection (negotiation or auction), procurement. S136
- Metrics: **days payable outstanding**, average purchase price, supply quality, supply lead time, on-time deliveries, price range, purchase quantity, supplier reliability. S137–S138
- Trade-off: **increase the supply chain surplus**; outsource only if it raises total surplus more than doing it in-house. S139

Pricing

- Sets revenue and shapes demand; short-term discounts can shift demand to match an inflexible supply. S140
- **Costco**: consistently low prices for a segment that values affordability over availability. S141
- **Amazon, July 2008**, two books worth $30: standard 3–5 days **$4.98**; two-day **$13.97**; one-day **$22.97**; free shipping 7–14 days. Free-shipping customers level out warehouse workload. S142
- Decisions: pricing and economies of scale (quantity discounts), **everyday low pricing vs high-low pricing**, **fixed vs menu pricing**. S143
- Metrics: profit margin, **days sales outstanding**, incremental fixed cost per order, incremental variable cost per unit, average sale price, average order size, range of sale price, range of periodic sales. S144–S146
- Trade-off: **increase firm profits**; EDLP stabilizes demand. S147

WRAP-UP S148

Financial measures: ROE, ROA, accounts payable turnover, profit margin, asset turnover, receivable turnover, inventory turns, PP&E turns, cash-to-cash cycle. Six drivers create **strategic fit** by balancing responsiveness and efficiency.

BEFORE THE EXAM ALL LECTURES

Cheat sheet

DRIVER	RESPONSIVE CHOICE	EFFICIENT CHOICE	TRADE-OFF
Facilities	Many, decentralized, spare capacity	Few, centralized, high utilization	Responsiveness vs efficiency
Inventory	Finished goods near customers	Centralized, raw form	Responsiveness vs efficiency
Transportation	Air, express	Sea, rail	Responsiveness vs efficiency
Information	Share the minimum needed		Complexity vs value
Sourcing	Local, in-house (Zara, Cisco high-end)	Low-cost country (Cisco low-end)	Increase SC surplus
Pricing	Menu pricing for speed	Everyday low pricing	Increase firm profits

Risk vs uncertainty	Risk can be estimated; uncertainty cannot S13
Macro processes	CRM, ISCM, SRM S20
SCOR	Plan, Source, Make, Deliver, Return · 4Ps · 250+ metrics S29–S34
4 cycles	Customer order, replenishment, manufacturing, procurement S39
Little's law	I = D × T S104
Numbers	37% inventory cut (weekly plans) · Intel $5.50→ –$4, 9→2 weeks · MILP –10%, 95% next-day · Sunsweet 30%→<15%

MEMORY HOOK (MY OWN, NOT FROM THE SLIDES)

Logistical drivers are the things you can touch: **F**acilities, **I**nventory, **T**ransportation ("FIT"). Cross-functional ones are decisions: **I**nformation, **S**ourcing, **P**ricing ("ISP").

READ ONCE SOURCE CHECK

Slide quirks to watch

- **SCOR dimensions:** S29 lists five (reliability, responsiveness, agility, costs, asset management); the metrics slide S34 lists only four, without agility.
- The "Source" line is missing from the five-process slide; only its tail ("products, and services from suppliers to support production") survives. S30
- The "Types of SC Models" graphic shows an "inventory model" and omits efficient and agile, while the slides that follow describe continuous, fast, efficient, agile, custom-configured, flexible and virtual. S61
- Intel's "cost reduction of more than $4 per unit" against an original $5.50 can be read as a cut *to* about $1.50 per chip. S82
- Facility metrics say unit costs decline "(PPET increases)" with utilization; PPET means property, plant and equipment turns. S101
- Video or link-only slides: S12, S79. S24 and S52 are reference lists.

SELF-TEST 50 QUESTIONS

Week 6 practice questions

Answer these first, then check the key on the next pages. Option letters match the online quiz.

Q1 LECTURE 26

Implied demand uncertainty refers to uncertainty in:

A the portion of demand the supply chain plans to satisfy, given its service requirements
B exchange rates
C supplier lead times
D total market demand only

Q2 LECTURE 26

If a supply chain raises its service level while underlying demand uncertainty stays the same, implied demand uncertainty:

A becomes zero
B increases
C stays the same
D decreases

Q3 LECTURE 26

Which firm faces higher implied demand uncertainty?

A One supplying the same product with a long lead time
B One serving mainly emergency orders
C Both equal
D Neither

Q4 LECTURE 26

Which product would have LOW supply uncertainty per the slides?

A A brand-new cell phone
B Salt sold in a supermarket
C A new PC component
D A prototype drone

Q5 LECTURE 26

Risk differs from uncertainty because risk:

A only affects suppliers
B is a function of outcome and probability, so it can be estimated
C only affects customers
D cannot be estimated

Q6 LECTURE 26

Increasing plan reviews from yearly to weekly can reduce inventory levels by about:

A 7%
B 37%
C 17%
D 73%

Q7 LECTURE 26

Which macro process focuses on the interface between the firm and its customers?

A CRM
B ISCM
C SRM
D ERP

Q8 LECTURE 26

Capacity planning, demand and supply planning and order fulfilment belong to which macro process?

A SCOR
B ISCM
C SRM
D CRM

Q9 LECTURE 26

Supplier evaluation and selection, and supply term negotiation, belong to:

A CRM
B ISCM
C PDCA
D SRM

Q10 LECTURE 26

Which category of uncertainty involves consignors and consignees, receivable risks and forecast errors?

A Information
B Customer-related
C Logistics
D Environment

Q11 LECTURE 27

The SCOR model was given by:

A Gartner
B World Bank
C Supply-Chain Council
D ISO

Q12 LECTURE 27

The five primary management processes of SCOR are:

A Plan, Source, Make, Deliver, Return
B Sort, Set, Shine, Standardize, Sustain
C Plan, Do, Check, Act, Review
D Define, Measure, Analyze, Improve, Control

Q13 LECTURE 27

Which is NOT one of the 4Ps (elements) of the SCOR model?

A Performance
B Process
C Price
D People

Q14 LECTURE 27

Roughly how many metrics come with the SCOR framework?

A About 50
B About 25
C More than 250
D Exactly 5

Q15 LECTURE 27

Which is listed as a DISADVANTAGE of SCOR?

A Whole supply chain scope
B Limited scope: little focus on social responsibility and sustainability
C Common repository of terms
D Enables benchmarking

Q16 LECTURE 27

In the cycle view, the replenishment cycle occurs at the interface between:

A customer and retailer
B distributor and manufacturer
C retailer and distributor
D manufacturer and supplier

Q17 LECTURE 27

Which cycle occurs between the manufacturer and the supplier?

A Procurement cycle
B Customer order cycle
C Replenishment cycle
D Manufacturing cycle

Q18 LECTURE 27

A push process is executed:

A only after payment
B only for returns
C in anticipation of customer orders (speculative)
D in response to a customer order (reactive)

Q19 LECTURE 27

In the L.L. Bean push/pull example, which cycle is a pull process?

A Customer order cycle
B Procurement cycle
C Manufacturing cycle
D Replenishment cycle

Q20 LECTURE 27

The cycle view of the supply chain is especially useful for:

A strategic decisions
B operational decisions
C pricing decisions
D marketing campaigns

Q21 LECTURE 28

Which is described as the most critical component of supply chain modelling because it directly affects the bottom line?

A Inventory
B IT
C Customer service
D Transportation

Q22 LECTURE 28

Which company is the slide's example of the continuous model?

A Nike
B Dell
C Zara
D Amazon

Q23 LECTURE 28

Dell offering individualized computer configurations illustrates which model?

- **A** Efficient
- **B** Virtual
- **C** Custom-configured
- **D** Continuous

Q24 LECTURE 28

Which is NOT one of the four characteristics an agile supply chain needs?

- **A** Market sensitivity
- **B** Lowest unit cost
- **C** Process alignment
- **D** Virtual integration

Q25 LECTURE 28

The three requirements for a flexible supply chain model are:

- **A** JIT, kanban, kaizen
- **B** plan, source, make
- **C** low cost, high volume, long lead time
- **D** part segmentation, precise stocking algorithms, flexible planning

Q26 LECTURE 28

In Zara's segmentation, standard and highly predictable items are sourced through:

- **A** Asia airfreight
- **B** near offshore (Turkey)
- **C** Asia sea freight (China, India)
- **D** local fast response (Spain, Portugal, Morocco)

Q27 LECTURE 28

A supply chain control tower is best described as:

- **A** an airport tower for cargo
- **B** a warehouse layout
- **C** a type of 4PL contract
- **D** a cloud-based solution using ML, AI and IoT for end-to-end, real-time visibility

Q28 LECTURE 28

In the Intel Atom case, the only lever available to reduce supply chain cost was:

- **A** packaging
- **B** duty payments
- **C** inventory
- **D** distribution

Q29 LECTURE 28

Intel reduced the Atom chip's order cycle time from:

- **A** two weeks to two days
- **B** four weeks to one week
- **C** nine weeks to two weeks
- **D** five days to two days

Q30 LECTURE 28

In the chemical company case, the MILP network model achieved:

- **A** 20% cost cut, two-day delivery
- **B** 5% cost cut, same-day delivery for all
- **C** no measurable change
- **D** 10% cost reduction and next-day delivery for 95% of products

Q31 LECTURE 29

Which three are the logistical drivers in Chopra & Meindl's framework?

- **A** Plan, source, make
- **B** Facilities, inventory, transportation
- **C** Cost, quality, time
- **D** Information, sourcing, pricing

Q32 LECTURE 29

Cross-functional drivers, compared with logistics drivers, mainly affect:

- **A** responsiveness and agility
- **B** efficiency and cost
- **C** fuel use
- **D** warehouse layout only

Q33 LECTURE 29

Centralizing facilities generally:

- **A** increases efficiency but reduces responsiveness
- **B** reduces efficiency but increases responsiveness
- **C** increases both efficiency and responsiveness
- **D** has no effect

Q34 LECTURE 29

Flow time efficiency is the ratio of:

- **A** batch size to demand
- **B** theoretical flow time to actual average flow time
- **C** capacity to utilization
- **D** actual flow time to theoretical flow time

Q35 LECTURE 29

By Little's law, if throughput is 100 units per day and flow time is 5 days, inventory is:

- **A** 95 units
- **B** 20 units
- **C** 500 units
- **D** 105 units

Q36 LECTURE 29

Inventory held as a buffer against unexpected demand or late deliveries is:

- **A** safety inventory
- **B** cycle inventory
- **C** obsolete inventory
- **D** seasonal inventory

Q37 LECTURE 29

Days of Inventory Outstanding (DIO) is calculated as:

- **A** COGS ÷ average inventory
- **B** average inventory ÷ COGS per day
- **C** units shipped ÷ days
- **D** sales ÷ receivables

Q38 LECTURE 29

Dell flying components from Asia is an example of using transportation to:

- **A** cut transport cost
- **B** avoid tariffs
- **C** use sea freight
- **D** reduce inventory and increase responsiveness at higher transport cost

Q39 LECTURE 29

Blue Nile combines ____ transportation with ____ inventory.

- **A** responsive (FedEx overnight); centralized
- **B** rail; regional
- **C** sea; local store
- **D** slow; decentralized

Q40 LECTURE 29

Outbound transportation costs mainly affect which line in the financials?

- **A** Fixed assets
- **B** COGS
- **C** Accounts payable
- **D** Selling, general and administrative expenses

Q41 LECTURE 30

Regarding information sharing, the slides advise supply chains to share:

- **A** all available information
- **B** no information with suppliers
- **C** the minimum information necessary to achieve coordination
- **D** only financial data

Q42 LECTURE 30

After implementing an S&OP suite, Sunsweet Growers' production overruns fell from 30% to:

- **A** under 15%
- **B** under 5%
- **C** 25%
- **D** 20%

Q43 LECTURE 30

The forecast horizon must be:

- **A** exactly one week
- **B** shorter than the decision lead time
- **C** greater than or equal to the lead time of the decision it drives
- **D** longer than a year

Q44 LECTURE 30

A ratio of demand variability to order variability below one potentially indicates:

- **A** a stable chain
- **B** perfect forecasting
- **C** excess capacity
- **D** the bullwhip effect

Q45 LECTURE 30

The overall trade-off for the information driver is:

- **A** speed vs safety
- **B** complexity vs value
- **C** responsiveness vs efficiency
- **D** cost vs quality

Q46 LECTURE 30

Cisco's sourcing for low-end home routers aims for:

- **A** responsiveness via US contract manufacturers
- **B** efficiency via production in China
- **C** air freight
- **D** in-house production

Q47 LECTURE 30

Days payable outstanding measures:

A days to deliver an order
B days of inventory on hand
C days to collect cash from customers
D days between a supplier performing a task and being paid

Q48 LECTURE 30

In July 2008, Amazon's one-day shipping for two books worth $30 cost:

A $22.97
B $13.97
C $4.98
D Free

ANSWERS 50 QUESTIONS

Answer key

Q	ANS	WHY · SLIDE
Q1	A	S5: it depends on lead times, order frequency and service levels. S5
Q2	B	S6: a higher service level means meeting rare surges too. S6
Q3	B	Long lead times let orders be filled evenly. S6
Q4	B	Mature products like salt; new products have high supply uncertainty. S8
Q5	B	Uncertainty: decision makers cannot estimate outcome or probability. S13
Q6	B	S17: 37%. S17
Q7	A	CRM: customers; ISCM: internal; SRM: suppliers. S20
Q8	B	ISCM aims to fulfil demand efficiently. S21
Q9	D	SRM. S21
Q10	B	S16. S16
Q11	C	A global non-profit consortium. S29
Q12	A	S29. S29
Q13	C	The 4Ps are Performance, Process, Practice, People. S32
Q14	C	S34. S34
Q15	B	Others: complexity, one-size-fits-all, inflexibility, data dependence. S36
Q16	C	Customer order: customer–retailer; replenishment: retailer–distributor; manufacturing: distributor–manufacturer; procurement: manufacturer–supplier. S39
Q17	A	S39. S39
Q18	C	Pull = reactive; push = speculative. S48
Q19	A	The customer order cycle is pull; the others are push. S49
Q20	B	S41: roles of each member are defined. S41
Q21	A	S59. S59
Q22	D	Zara → fast; Dell → custom-configured; Nike → agile; HP → flexible; P&G → virtual. S63
Q23	C	S71. S71
Q24	B	Four traits: process alignment, virtual integration, network base, market sensitivity. S70
Q25	D	S72; HP is the example. S72
Q26	C	High-fashion items are made locally for fast response. S68
Q27	D	S78. S78
Q28	C	Intel moved to make-to-order to cut the nine-week order cycle. S80
Q29	C	S82; cost fell by more than $4 per unit. S82
Q30	D	MILP = mixed-integer linear programming. S84
Q31	B	Cross-functional drivers: information, sourcing, pricing. S95
Q32	A	S97. S97
Q33	A	Decentralizing is the reverse. S98
Q34	B	Low values mean much of the time is spent waiting. S102
Q35	C	I = D × T = 100 × 5 = 500. Little's law links inventory, flow time and throughput. S104
Q36	A	S107. S107
Q37	B	Inventory turnover is COGS (or sales) ÷ average inventory. S109
Q38	D	S113. S113
Q39	A	No physical stores, lower costs. S114
Q40	D	Inbound transport is part of COGS; outbound affects SG&A. S115
Q41	C	Beyond a point marginal cost exceeds marginal benefit. S127
Q42	A	Forecast accuracy also improved 15–20%; alerts came 2–3 weeks earlier. S128
Q43	C	S130. S130
Q44	D	S131. S131
Q45	B	Sourcing: increase SC surplus; pricing: increase firm profits. S132
Q46	B	High-end products go to responsive US contract manufacturers. S135
Q47	D	Days sales outstanding (S144) is the customer-side measure. S137
Q48	A	Standard $4.98, two-day $13.97, one-day $22.97, free in 7–14 days. S142
Q49	A	EDLP leads to stable demand. S143

Q	ANS	WHY · SLIDE
Q50	D	S141. S141

Made from *WEEK 6.pdf* (150 slides, NPTEL Logistics & Supply Chain Management, Prof. Vikas Thakur, IIT Kharagpur). Notes use only the slides, except boxes marked "my own".